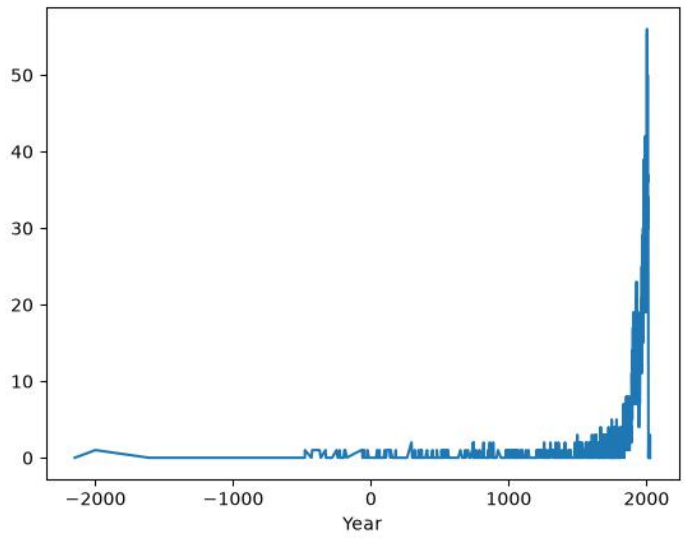
1.1 加载-定位-计算-排序-打印

```
[27]: Country
      CHINA           2139209.0
      TURKEY          1199720.0
      IRAN            1014453.0
      ITALY            492236.0
      SYRIA            419826.0
      HAITI            323484.0
      AZERBAIJAN       319251.0
      JAPAN            242516.0
      ARMENIA          177890.0
      PAKISTAN         145085.0
      IRAQ             136200.0
      ECUADOR          135496.0
      TURKMENISTAN     117412.0
      PERU             102309.0
      ISRAEL            93388.0
      PORTUGAL          83572.0
      GREECE            81162.0
      CHILE             64277.0
      INDIA             63507.0
      TAIWAN            57154.0
      Name: Deaths, dtype: float64
```

1.2 先找到 Ms 值大于 3.0 的所有行，然后将这些行按年份分组并求和，最后画图展示

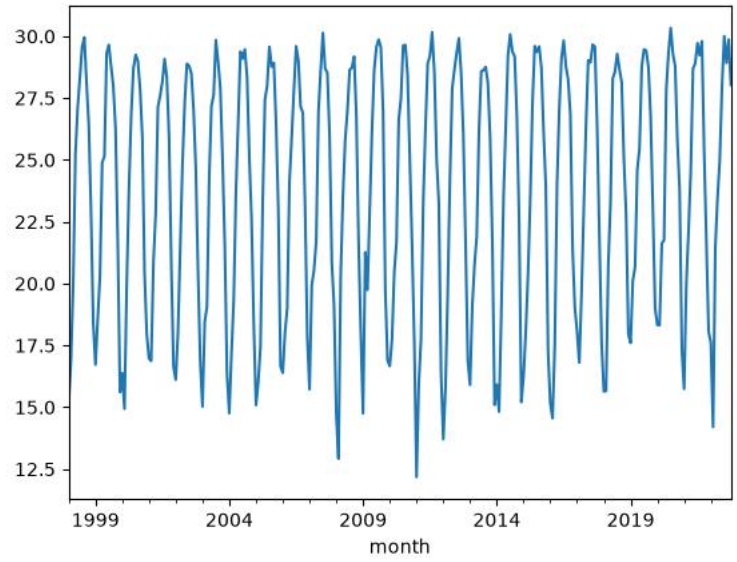


1.3 先按国家分组并得到一个新的表格-识别 MS 空白值-找到最大 MS-识别日期-输出

Country	TotalEq	LargestEq
CHINA	626	1920/12/16, CHINA: GANSU PROVINCE, SHANXI PROVINCE
JAPAN	429	869/7/13, JAPAN: SANRIKU
INDONESIA	427	2004/12/26, INDONESIA: SUMATRA: ACEH: OFF WEST COAST
IRAN	385	856/12/22, IRAN: DAMGHAN, QUMIS
TURKEY	359	1939/12/26, TURKEY: ERZINCAN
ITALY	338	1915/1/13, ITALY: MARSICA, AVEZZANO, ABRUZZI
GREECE	296	365/7/21, GREECE: CRETE: KNOSSOS
USA	283	1957/3/9, ALASKA
PHILIPPINES	233	1897/9/21, PHILIPPINES: MINDANAO, ZAMBOANGA, SULU, ISABELA
MEXICO	217	1787/3/28, MEXICO: SAN MARCOS, OAXACA
CHILE	200	1730/7/8, CHILE: VALPARAISO
PERU	195	1619/2/14, PERU: TRUJILLO, PIURA, SANTA
RUSSIA	159	1952/11/4, RUSSIA: KAMCHATKA PENINSULA
PAPUA NEW GUINEA	107	2000/11/16, PAPUA NEW GUINEA: NEW IRELAND, DUKE OF YORK
TAIWAN	103	1920/6/5, TAIWAN
INDIA	102	1897/6/12, INDIA: ASSAM; BANGLADESH
COLOMBIA	83	1904/1/20, PANAMA-COLOMBIA

2 加载文件-识别日期-将 TMP 列的所有值转为字符串并识别是否为无效值，同时去除后缀-生成没有无效值的表格-转会数值并除以缩放因子变成实际值-画图展示

根据说明文件+9999,9 记为无效值。，9 代表无效，+9999 在实际中也不可能出现。在使用的数据集中仅出现+9999,9 这一种无效值样式



3

```
usecols=range(17),#只读取前 17 列
skiprows=[1, 2], #跳过文件中的第 2、3 行
parse_dates=['ISO_TIME'],#把 ISO_TIME 列解析成 datetime，后续可以直接 .dt.year 等
na_values=['NOT_NAMED', 'NAME'])#指定哪些字符串视为缺失值
```

3.1 分组-整合-找最大-排序-展示

```
[8]: SID
     1886225N13320    NaN
     1886226N15336    NaN
     1886277N14279    NaN
     1887226N14328    NaN
     1887226N16302    NaN
     1888228N23289    NaN
     1888239N13328    NaN
     1888283N22267    NaN
     1889255N16301    NaN
     1890237N13311    NaN
     Name: NAME, dtype: str
```

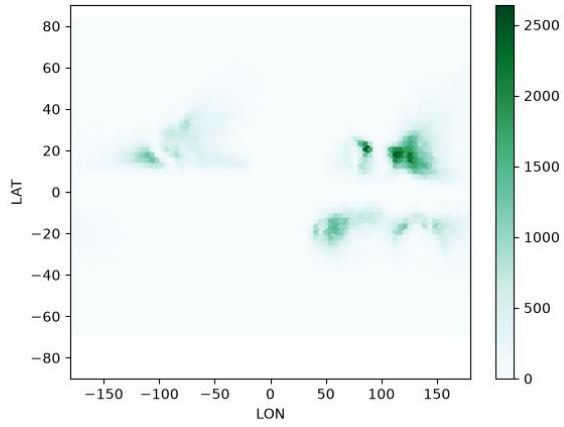
3.2 分组-找最大-排序-展示

```
[9]: SID
     1886225N13320    95
     1886226N15336    95
     1886277N14279    95
     1887226N14328    95
     1887226N16302    95
     1888228N23289    95
     1888239N13328    95
     1888283N22267    95
     1889255N16301    95
     1890237N13311    95
     1891229N11313    95
     1893225N12331    95
     1893226N10310    95
     1893228N18340    95
     1893268N12335    95
     1893271N16278    95
     1894242N13325    95
     1894261N13311    95
     1894270N12284    95
     1894284N11303    95
     Name: WMO_WIND, dtype: str
```

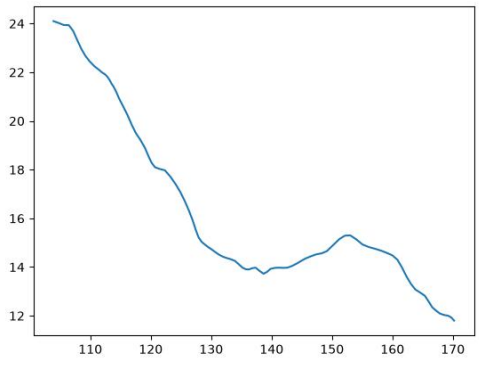
3.3 先按流域分组，然后求各自数据点总数量，最后排序

```
[12]: BASIN
      WP    236576
      SI    160668
      SP     67119
      EP     62412
      NI     55401
      SA       119
      dtype: int64
```

3.4 直接画图



3.5 先按名字和时间定位找到目标台风-按日期排序-按经纬度画图



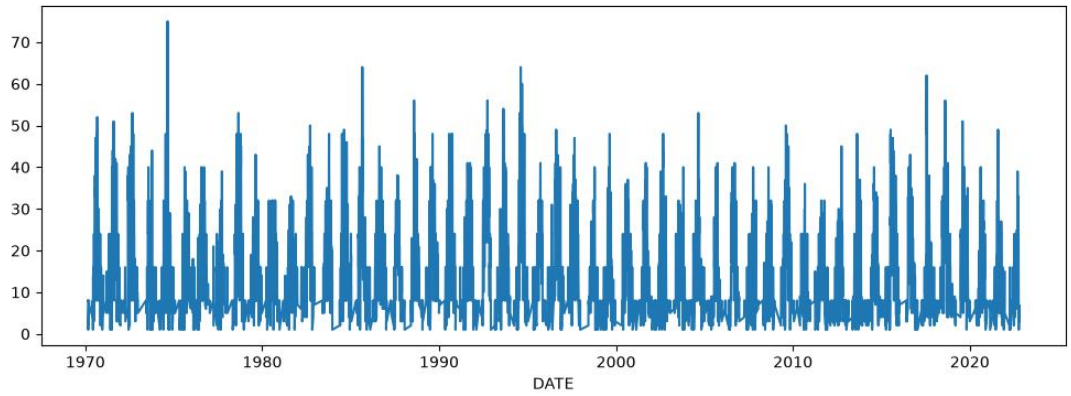
3.6 按所给条件定位

[18]:

	SID	SEASON	NUMBER	BASIN	SUBBASIN	NAME	ISO_TIME	NATURE	LAT	LON	WMO_WIND	WMO_PRES	WMO_AG
350393	1970050N07151	1970	22	WP	MM	NANCY	1970-02-19 00:00:00	TS	7.00000	151.400		1006	
350394	1970050N07151	1970	22	WP	MM	NANCY	1970-02-19 03:00:00	TS	7.24752	151.205			
350395	1970050N07151	1970	22	WP	MM	NANCY	1970-02-19 06:00:00	TS	7.50000	151.000		1002	
350396	1970050N07151	1970	22	WP	MM	NANCY	1970-02-19 09:00:00	TS	7.75747	150.772			
350397	1970050N07151	1970	22	WP	MM	NANCY	1970-02-19 12:00:00	TS	8.00000	150.500		998	

3.7 分组-汇总-画图

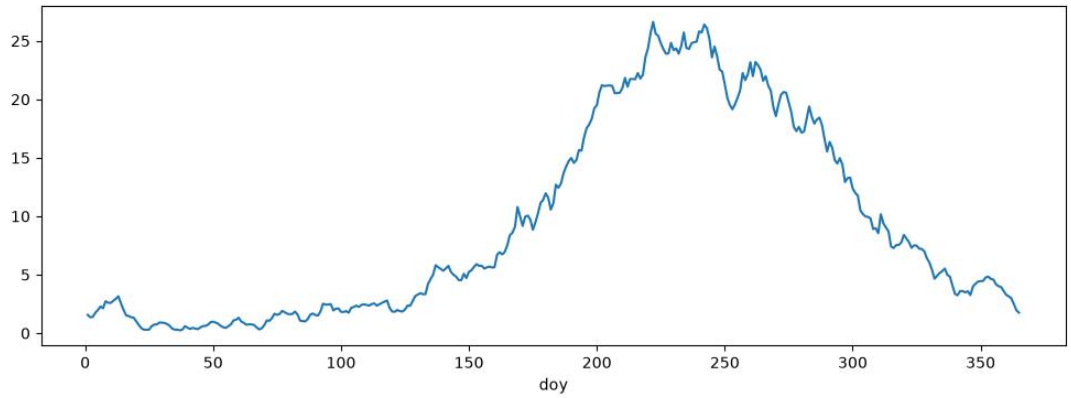
3.8 排除闰年的特殊统一按 365 天处理




```
#3.8排除了闰年影响
doy = df2['ISO_TIME'].dt.dayofyear.rename('doy')
yr  = df2['ISO_TIME'].dt.year.rename('year')
need3_8 = df2.groupby([yr, doy]).size().unstack(fill_value=0)

clim = need3_8.loc[:, need3_8.columns <= 365].mean(axis=0).sort_index()

clim.plot(figsize=(12, 4))#拉长画布
plt.show()
```

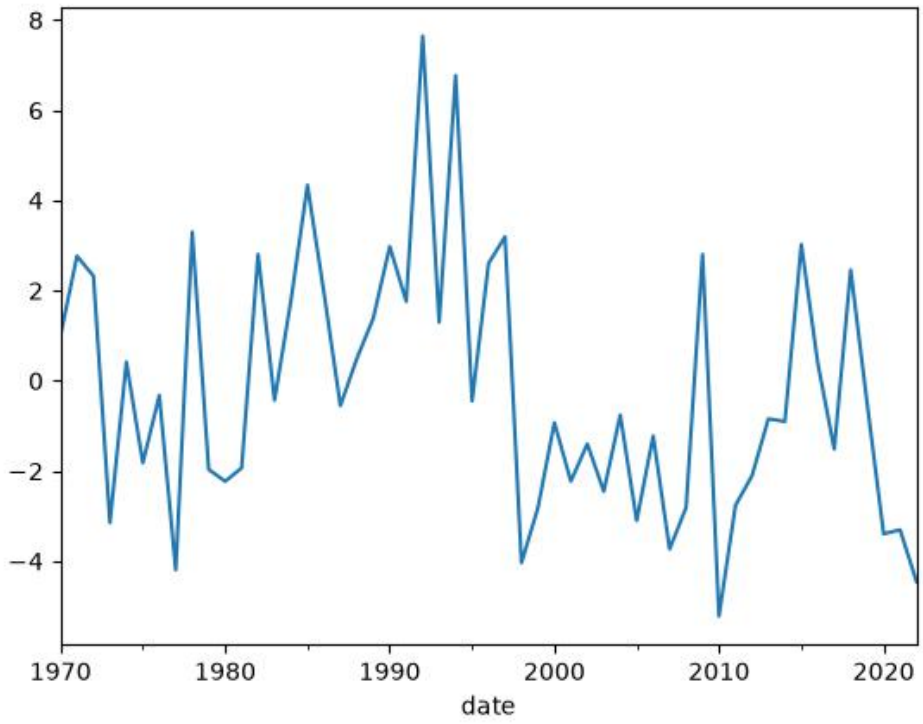


3.9 每天的减去平均值-展示

[24]:

	year	doy	count	clim	anomaly
0	1970	50	8	6.500000	1.500000
1	1970	51	8	8.000000	0.000000
2	1970	52	8	7.166667	0.833333
3	1970	53	8	6.600000	1.400000
4	1970	54	8	5.400000	2.600000

3.10




```
正异常最大的年份（飓风偏活跃）：
date
1992-12-31    7.645086
1994-12-31    6.769985
1985-12-31    4.338109
1978-12-31    3.298956
1997-12-31    3.195031
Name: anomaly, dtype: float64

负异常最大的年份（飓风偏不活跃）：
date
2010-12-31   -5.217217
2022-12-31   -4.456293
1977-12-31   -4.192921
1998-12-31   -4.035779
2007-12-31   -3.731739
Name: anomaly, dtype: float64
```

4 使用了来自美国国家气候数据中心（National Climatic Data Center of the United States，NCDC）的广州市 2024 年的气象数据，对其中逐小时的气温数据作分析

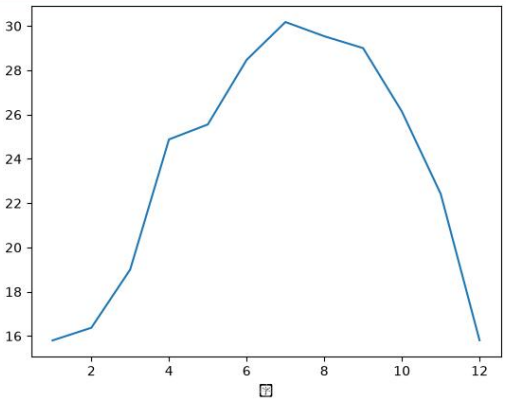
偏度-0.61 分布左侧拖了一条尾巴，存在少量明显偏低的样本。整体波动偏小。整体平均水平偏暖，接近"舒适—炎热"区间

[32]:
```
#4.1
t_gz = '气温（*10）'
gz = gz.loc[gz[t_gz] != -9999].copy()
```

[33]:
```
#4.2
gz['TEMP'] = gz[t_gz] / 10.0#除以缩放因子变成实际值

gz.groupby('月')['TEMP'].mean().sort_index().plot()
plt.show()
```

```
D:\tool\Anaconda\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 26376 (\N{CJK UNIFIED IDEOGRAPH-6708}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```



[35]:
```
last = {
    'mean':     gz['TEMP'].mean(),
    'median':   gz['TEMP'].median(),
    'var':      gz['TEMP'].var(),       # 方差
    'std':      gz['TEMP'].std(),
    'skew':     gz['TEMP'].skew(),      # 偏度
    'IQR':      gz['TEMP'].quantile(0.75) - gz['TEMP'].quantile(0.25),
}
print(pd.Series(last).round(2))
```

```
mean      23.67
median    25.00
var       45.77
std        6.77
skew      -0.61
IQR        8.00
dtype: float64
```